

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

May 2024

Course Code: INT405R01

Course: MACHINE LEARNING TECHNIQUES

QP No. :U311-6

Duration: 3 hours

Max. Marks:100

PART - A

Answer any Four questions

4 x 20 = 80 Marks

1. a) Imagine you are an intern in a startup company. The technical manager of the company has given you a data set capturing the number of new recruits inducted in the company for the past four years. The data is as given below.

S.No	Year (X)	Number of New Recruits (Y)
1	1	3
2	2	4
3	3	2
4	4	5

The manager is interested in predicting the number of people to be recruited for the next year based on this data. Compute and estimate the parameters of a simple line $\beta^T x + \beta_0 = 0$ that best fits these data points using Gradient descent based numerical univariate linear regression techniques. Assume a Learning rate of 0.01. Initial weight parameters as $\beta = 0$ and $\beta_0 = 1$. Update the parameters for two iterations. (15)

- b) Imagine after training, your Machine Learning model on the available data, you notice that it performs well on the training data

but poorly on new, unseen data. What will be your next course of action to improve this scenario. Justify your action with relevant technical arguments. (5)

2. a) A Valve can be used at four temperature levels. If the Valve is used at a cold temperature, then there is a probability of 0.003 that it will leak. If the Valve is used at a medium temperature, then there is a probability of 0.009 that it will leak. If the valve is used at a warm temperature, then there is a probability of 0.014 that it will leak. If the Valve is used at a hot temperature, then there is a probability of 0.018 that it will leak. Under standard operating conditions, the valve is used at a cold temperature 12% of the time, at a medium temperature 55 % of the time, at a warm temperature 20 % of the time and at a hot temperature 13% of the time.
- i) If the Valve leaks, what is the probability that it is being used at the hot temperature?
 - ii) If the Valve does not leak, what is the probability that it is being used at the medium temperature? (12)
- b) Consider an independent identically distributed data set and assume the data set is generated by a Gaussian Distribution. Justify with relevant mathematical expression, how will you identify the parameters of the Gaussian Distribution using Maximum Likelihood density Estimators. (8)
3. a) Imagine you are working on a food processing industry. You need to design a simple perceptron network for classifying apples from oranges. There are basically two features and you need to solve for the final classifying weights and biases for the given test data and initial weights and bias assumptions.

$$\left\{ \mathbf{p}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, t_1 = 0 \right\} \left\{ \mathbf{p}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, t_2 = 1 \right\} \left\{ \mathbf{p}_3 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}, t_3 = 0 \right\} \left\{ \mathbf{p}_4 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, t_4 = 1 \right\}$$

Use the initial weights and bias:

$$\mathbf{W}(0) = \begin{bmatrix} 0 & 0 \end{bmatrix} \quad b(0) = 0.$$

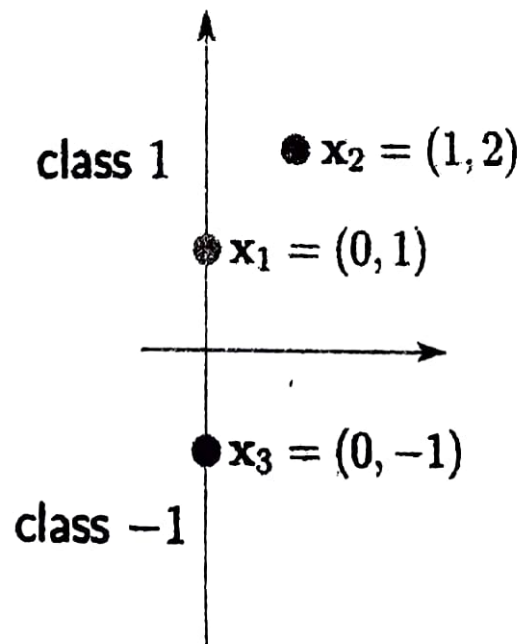
(15)

- b) Justify with relevant Mathematical expression, how Naïve Bayes Based Machine Learning algorithm is different from LDA and QDA based approach. (5)

4. a) For the data points as shown in below figure, compute the relevant linear hyperplane equation separating the two classes using Lagrangian based hard margin SVM using dual variables. (12)

Data Points

Input Data (X)	Output class (Y)
(1,2)	+1
(0,1)	+1
(0,-1)	-1



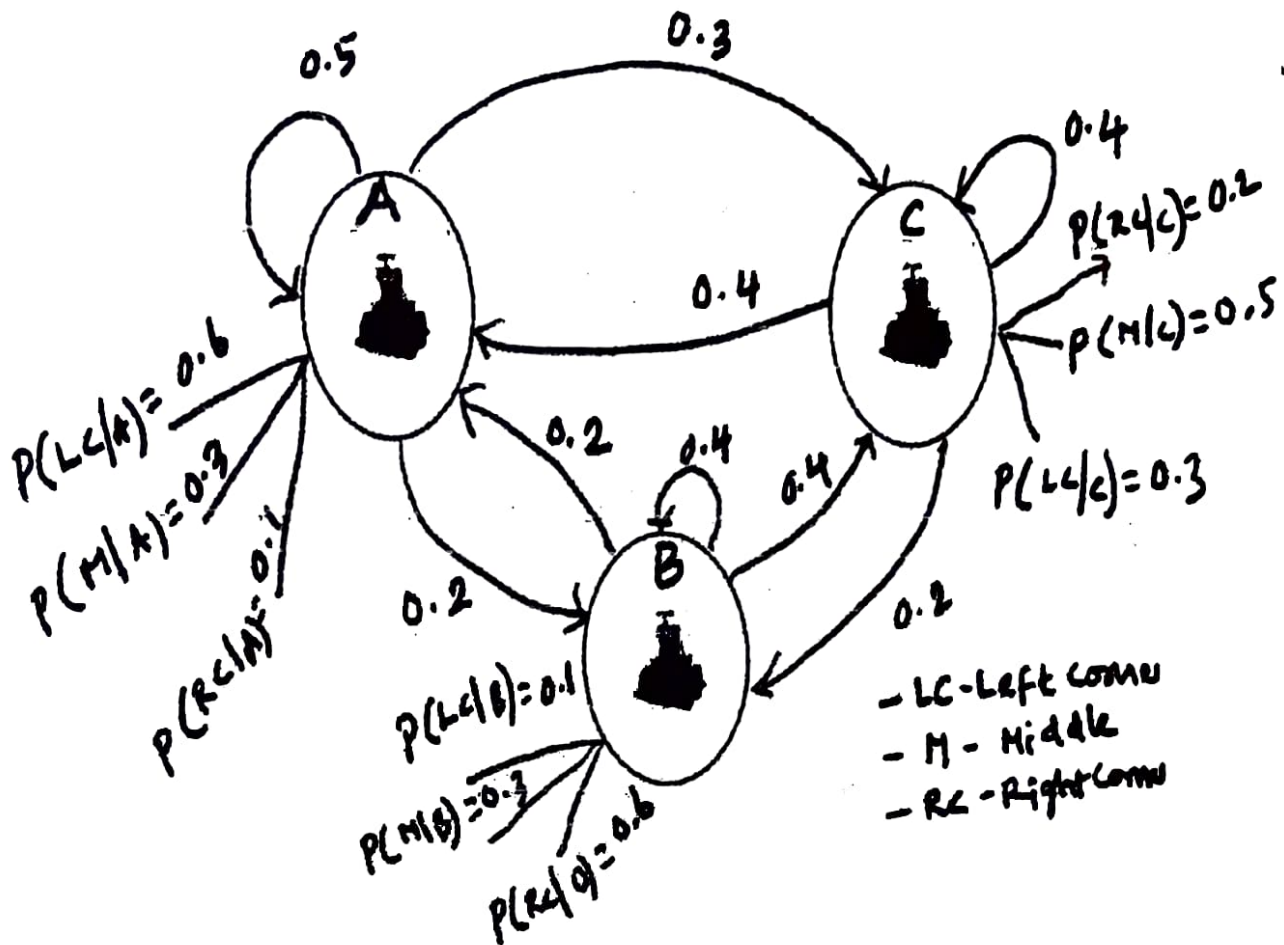
Imagine you are assigned the task of training a deep neural network to classify images into different categories. Despite achieving high accuracy on the training data set, you notice that the model performs poorly on the validation dataset, indicating overfitting. To address this problem, you decided to perform both data augmentation and drop outs. Justify with mathematical and architectural insights how these methods will solve this problem. (8)

- a) Compute the total number of parameters used in the CNN architecture given below: (8)

Input Layer	Images of Size: 32 x32 x3
Convolutional Layer 1	6 Filters of size 5x 5, Stride=1, Zero padding
Max Pooling Layer 1	2 x 2 Pooling with Stride 2
Convolutional Layer 2	16 Filters of size 5 x 5, Stride=1, Zero Padding
Max Pooling Layer 2	2 x 2 Pooling with stride 2
Fully Connected Layer 1	120 Neurons
Fully Connected Layer 2	84 Neurons
Output Layer	10 Neurons (10 Classes)

- b) Consider an LSTM network designed for a sequence-to-sequence prediction task, where the input is a sequence of words, and the output is a predicted sequence of words (Machine Translation). Explain how the LSTM network handles the input sequence to generate the output sequence, focusing on the flow of information through the LSTM layers. (12)
6. Consider a Mobile Robot which can be in any one of the three states (A, B, C) and at each state it's onboard sensor can give information about whether it is in Left Corner (LC), Middle(M) and Right Corner (RC) with an emission or observation probability as shown in the figure. The robot can change states and the Markovian transition probabilities are also as shown below. Compute the probability that

the model generated for a particular observation like (Left corner, left corner, Middle) using the Forward algorithm of HMM and draw the trellis diagram.



PART - B

Answer the following

1 x 20 = 20 Marks

- Imagine you are working in a Manufacturing Plant and you are designing a Neural network as shown in figure. Two neurons in the input side corresponds to the normalized Temperature and humidity readings. The hidden layer neurons and output layer neurons uses sigmoidal activation functions. The output neurons correspond to the predicted calibration adjustments needed for the heating and humidification systems. Train this Neural network with the initial set of weights, biases as shown in figure and assume a learning rate of

0.6, train it for one full iteration and correspondingly update the weights and biases.

